

UNSUPERVISED LEARNING

Clustering Analysis and Bayesian Network

Hidden patient sub-groups for mental-health outcomes
on the Kaggle Depression Dataset (413,768 rows).

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Can socio-economic and lifestyle factors reveal hidden patient groups linked to mental-health outcomes?

Six algorithms

- K-Means++,
- K-Prototypes,
- K-Modes,
- Hierarchical/Gower,
- DBSCAN,
- GMM
- A discrete Bayesian Network learned by Hill-Climbing, parametrised with BDeu, and explained with SHAP.

Problem & research goals

Why this matters?

Depression is among the most disabling mental-health conditions worldwide, with strong but **indirect** ties to socio-demographic, lifestyle and physiological variables.

Two complementary goals

1. Identify latent structure informative about **History of Mental Illness** (and, secondarily, Family History of Depression).
2. Encode probabilistic dependencies in a **Bayesian Network** that answers conditional queries under arbitrary partial evidence.

Methodological contribution

Six clustering algorithms are run on three parallel views of the data (numerical, mixed, fully-categorical) tailored to each algorithm's assumptions, then compared head-to-head against the Bayesian Network on the same supervised metrics.

Kaggle Depression Dataset

413,768

rows (adult respondents)

15

working variables (Name dropped)

3 / 5 / 3 / 4

numerical / ordinal / nominal / binary

30.4 %

positive class — History of Mental Illness

9,900

stratified, cleaned sub-sample used

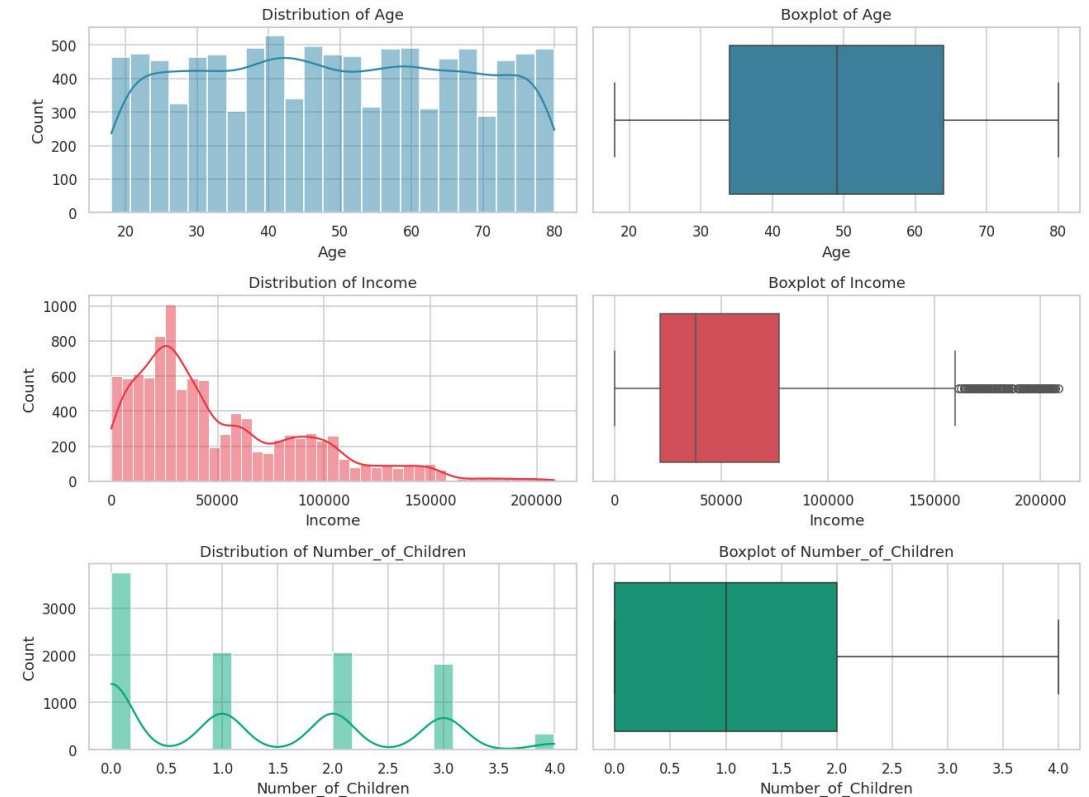


Fig. 1 — Age uniform [18, 80]; Income right-skewed (log1p before scaling); Number_of_Children zero-inflated.

Type-aware encoding · three parallel views

IQR filter (Income p0.5–p99.5, Age 18–90) removes only 1 % of rows. Income → log1p, numerals standardised, ordinals integer-coded, nominals one-hot, numerals quartile-binned for the categorical view.

X_num

Numerical / ordinal only

Standardised numerics + integer-coded ordinals.
Distance: Euclidean.

USED BY

K-Means++ · GMM · DBSCAN

X_mix

Mixed (numerical + categorical)

Numerics scaled, categoricals kept raw.
Distance: Gower (type-aware).

USED BY

K-Prototypes · Hierarchical

X_cat

Fully categorical

Quartile-binned numerics + label-encoded categoricals.
Distance: Hamming / matching.

USED BY

K-Modes · Bayesian Network

Architecture · end-to-end flow



Type-aware preprocessing feeds three parallel views; clustering and the Bayesian Network are compared head-to-head and unified through SHAP on a surrogate.

Manifold visualisation & cluster tendency

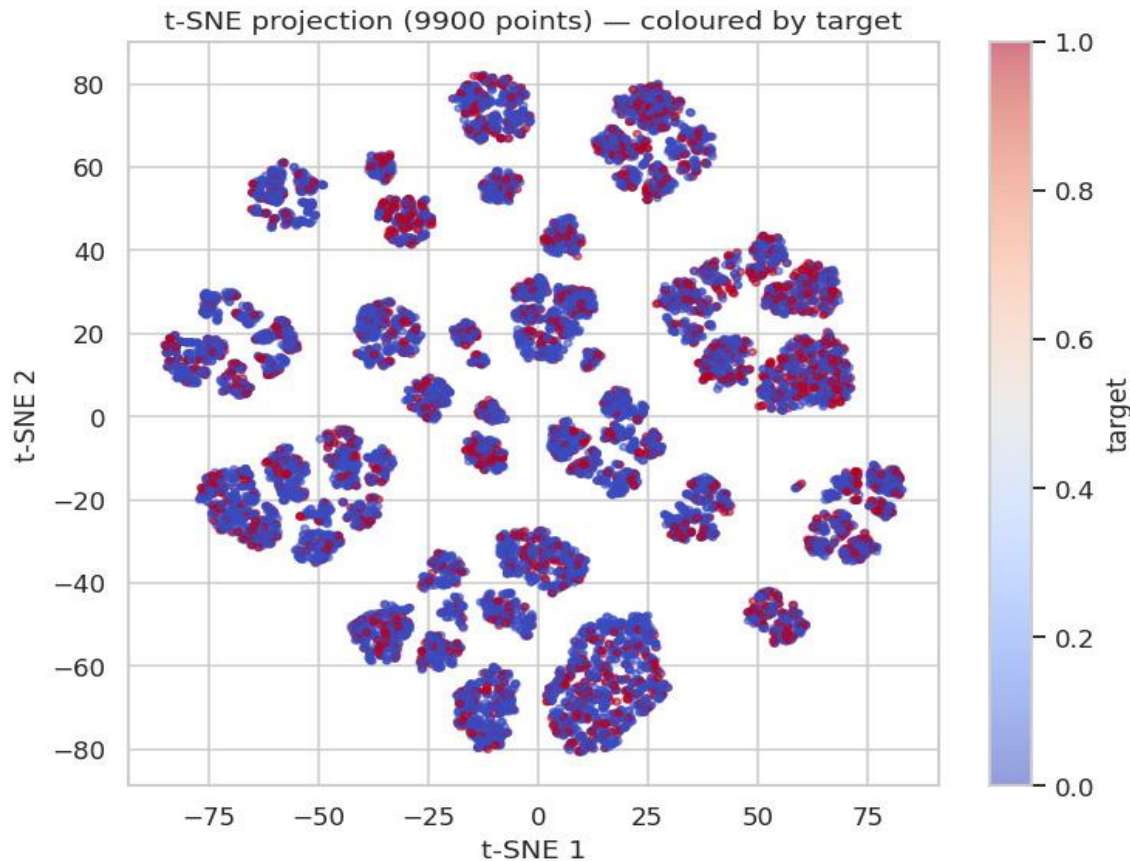


Fig. 3 — t-SNE projection (perplexity = 40) coloured by target.

Hopkins $H = 0.76$

Strong cluster tendency — far from a uniform distribution.

But the target does not align with the manifold

Red and blue points are mixed inside every t-SNE blob: the lifestyle / demographic clusters exist, yet they are only weakly informative about mental-illness status.

PCA: 2 components capture only 27 % of variance — 15 needed to reach 80 %. PCA is therefore not used as a preprocessing step.

Six Algorithms · Every Family

One technique per clustering paradigm

K-Means++

Partitioning · Numerical

K = 5

K-Prototypes

Partitioning · Mixed

K = 5

K-Modes

Partitioning · Categorical

K = 6

Hierarchical

Gower + Complete Linkage

K = 3

DBSCAN

Density-based

8 clusters · 2.3% noise

Gaussian Mixture

Probabilistic · EM + BIC

K = 10

K-Means++

Partitioning that minimises within-cluster squared distance

IDEA

Iteratively assigns points to the nearest centroid and recomputes means. K-Means++ seeding spreads initial centres to avoid poor local optima.

SETUP

View: X_num (standardised numerics + ordinals)
 Distance: Euclidean
 Init: k-means++ · n_init = 10
 K selection: silhouette + Calinski-Harabasz + elbow
 Chosen K = 5

RESULT

F1-macro = 0.41 · Purity 0.696 · Most balanced cluster sizes of all six methods. Strong agreement with GMM (FM = 0.70).

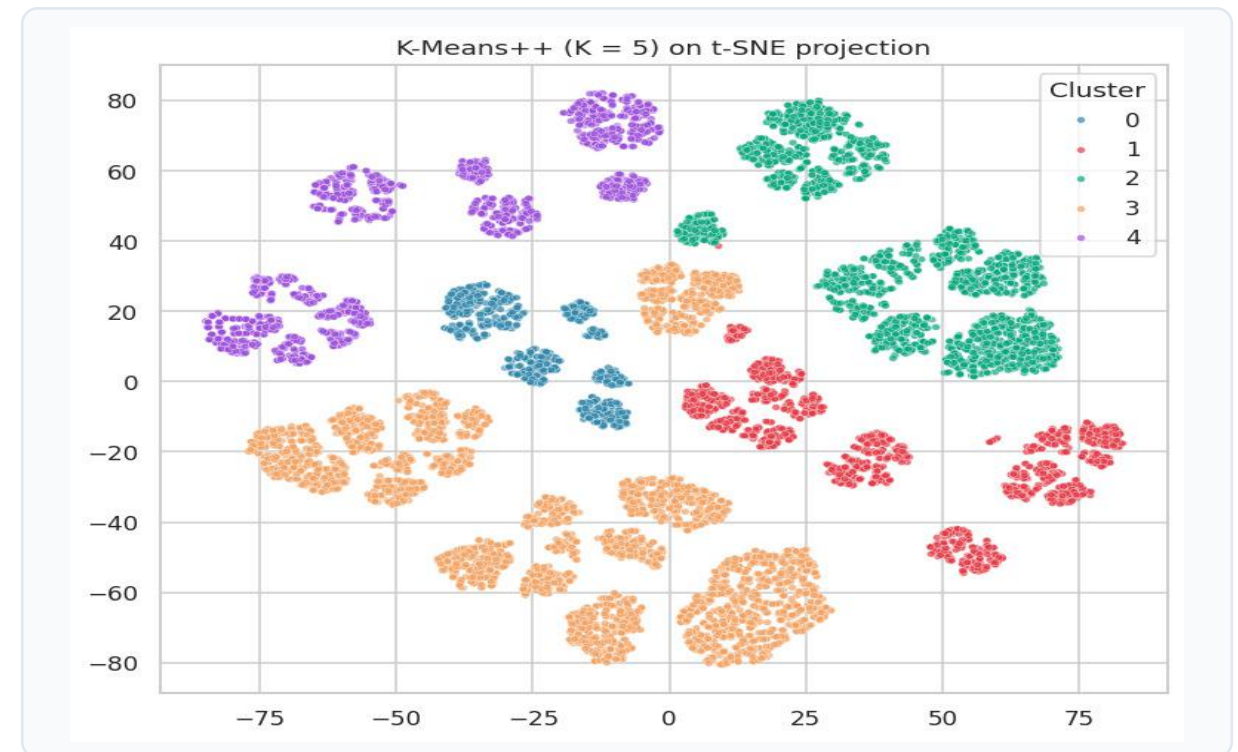


Fig. — K-Means++ (K = 5) projected on t-SNE: five well-separated blobs in the standardised numerical view.

K-Prototypes

K-Means extended to mixed numerical + categorical data

IDEA

Combines Euclidean distance on numerical features with a simple-matching dissimilarity on categoricals, weighted by γ . Centroids are means for numerics and modes for categoricals.

SETUP

View: `X_mix` (scaled numerics + raw categoricals)

γ balances numeric vs. categorical cost

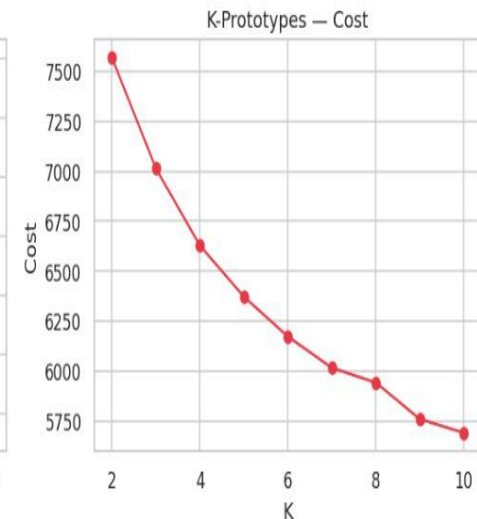
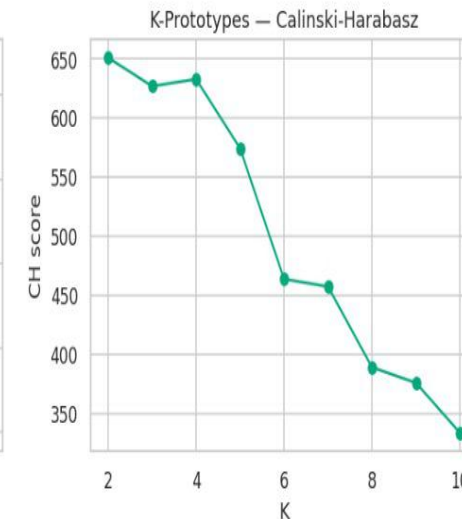
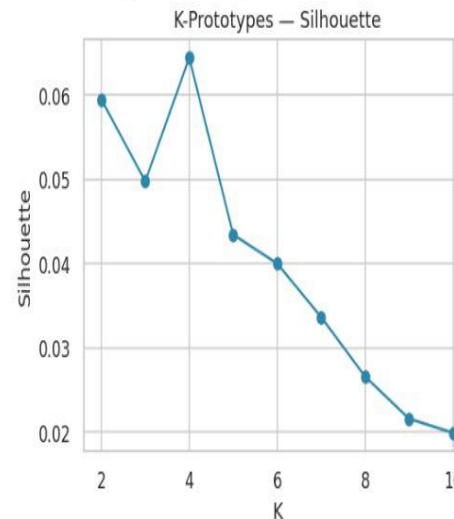
Init: `Cao · n_init = 10`

K selection: cost-elbow + silhouette

Chosen $K = 5$

RESULT

F1-macro = 0.433 — the best supervised score of all six clusterings. Purity 0.696 · balanced cluster sizes.



No dedicated figure in the report — see Fig. 5 (centroid family) for projection.

K-Modes

Mode-based partitioning on the fully-categorical view

IDEA

Replaces means with modes and Euclidean distance with simple-matching dissimilarity, so the optimisation lives entirely on categorical attributes.

SETUP

View: X_cat (quartile-binned numerics + label encoding)

Distance: Hamming / matching

Init: Cao

Validation: Silhouette · CH · cost-elbow

Chosen K = 6

RESULT

F1-macro = 0.41 · Purity 0.696. Pure-categorical clustering reaches the same ceiling as the centroid methods.

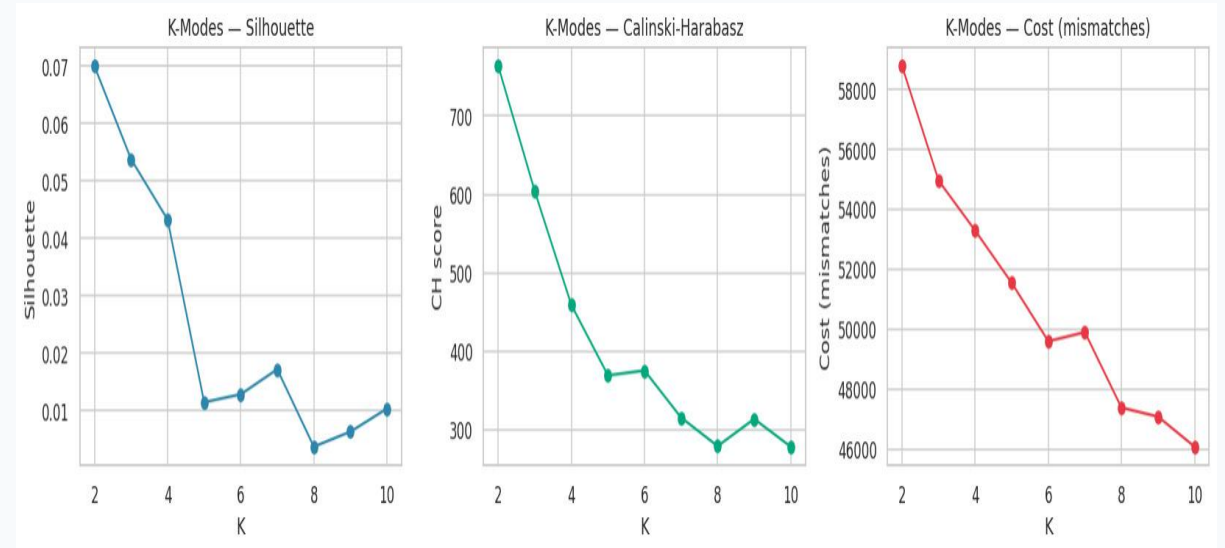


Fig. — Silhouette peaks at K = 2; the cost-curve elbow at K = 6 wins as majority-vote optimum.

Hierarchical · Gower

Agglomerative clustering on a Gower dissimilarity matrix

IDEA

Builds a full Gower dissimilarity (range 0.001–0.959) and merges clusters with complete linkage, less sensitive to noise than single linkage.

SETUP

View: X_mix · Distance: Gower (type-aware)

Linkage: complete

Learned on a 2,000-point sub-sample ($O(N^2)$ cost)

Labels propagated to 9,900 points via 1-NN

Chosen $K = 3$ (silhouette 0.057)

RESULT

F1-macro = 0.411 · Purity 0.696. Most skewed cluster sizes — structurally different partition from the Euclidean family ($FM < 0.35$).

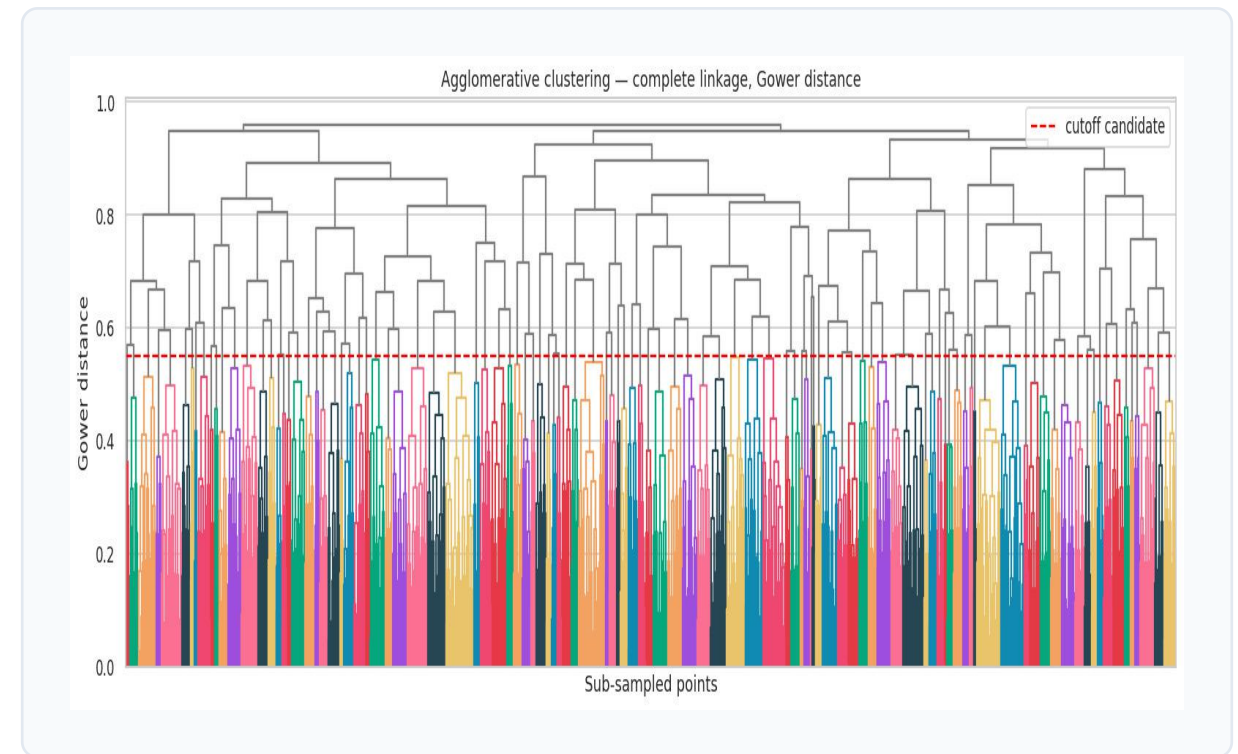


Fig. — Complete-linkage dendrogram on the 2,000-point Gower matrix. Red line marks the $K = 3$ cutoff.

DBSCAN

Density-based discovery of dense connected regions

IDEA

Groups points reachable through dense neighbourhoods and labels low-density points as noise. K is discovered automatically — no need to fix it in advance.

SETUP

View: X_num · Distance: Euclidean

$MinPts = \max(5, 2 \cdot \log n) = 18$

eps from automatic knee of the k -distance plot (≈ 3.17)

No manual K selection

RESULT

8 clusters discovered · only 2.3 % flagged as noise · Silhouette 0.147 ($\approx$ K-Means++) · F1-macro = 0.272 — lowest supervised score.

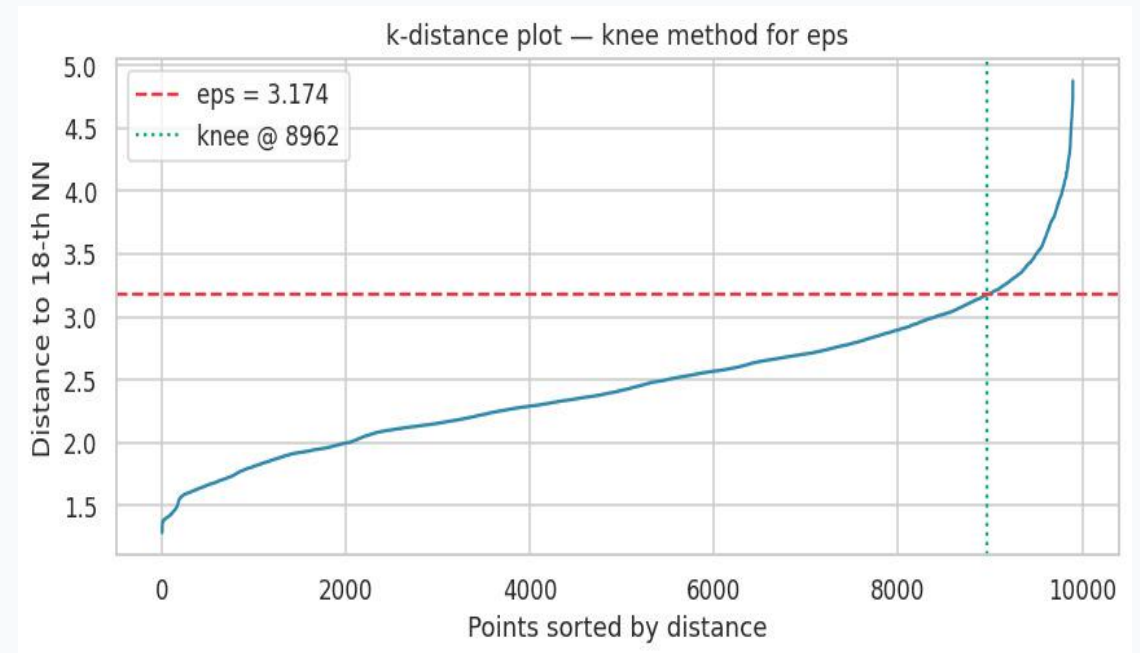


Fig. — k -distance plot; convex KneeLocator gives $eps \approx 3.17$ at index 8,962.

Gaussian Mixture Model

Soft probabilistic extension of K-Means

IDEA

Each cluster is modelled as a full-covariance Gaussian; points have soft membership probabilities estimated by Expectation–Maximisation.

SETUP

View: X_num · Covariance: full

Validation: $BIC = -2 \log L + p \cdot \log n$

Also tracked: AIC and silhouette

Chosen $K = 10$ (boundary)

RESULT

F1-macro = 0.41 · Purity 0.696. High Fowlkes–Mallows with K-Means++ (0.70) — the same Euclidean view drives similar partitions.

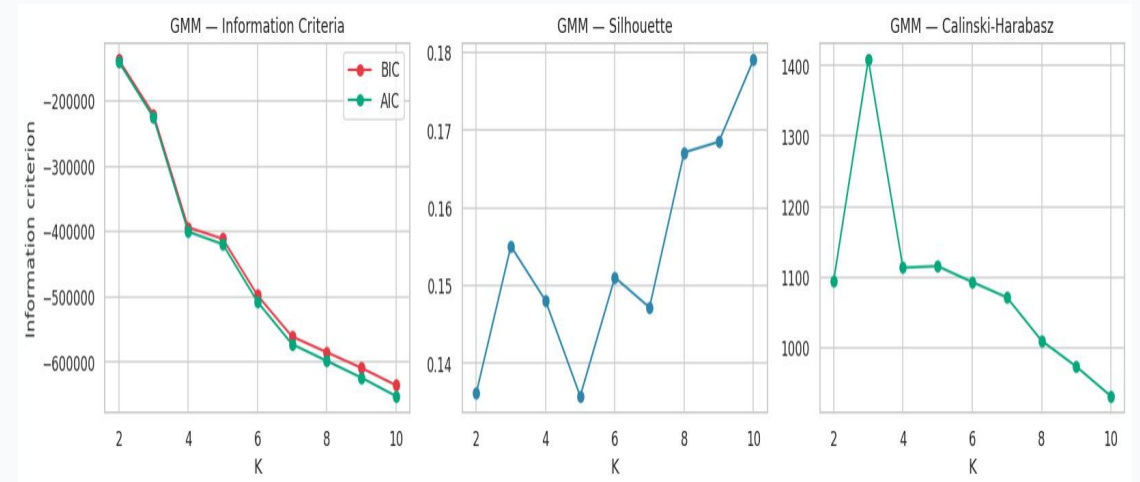


Fig. — BIC/AIC still decreasing at $K = 10$: no natural Gaussian cluster count in the data.

Pairwise agreement between clusterings

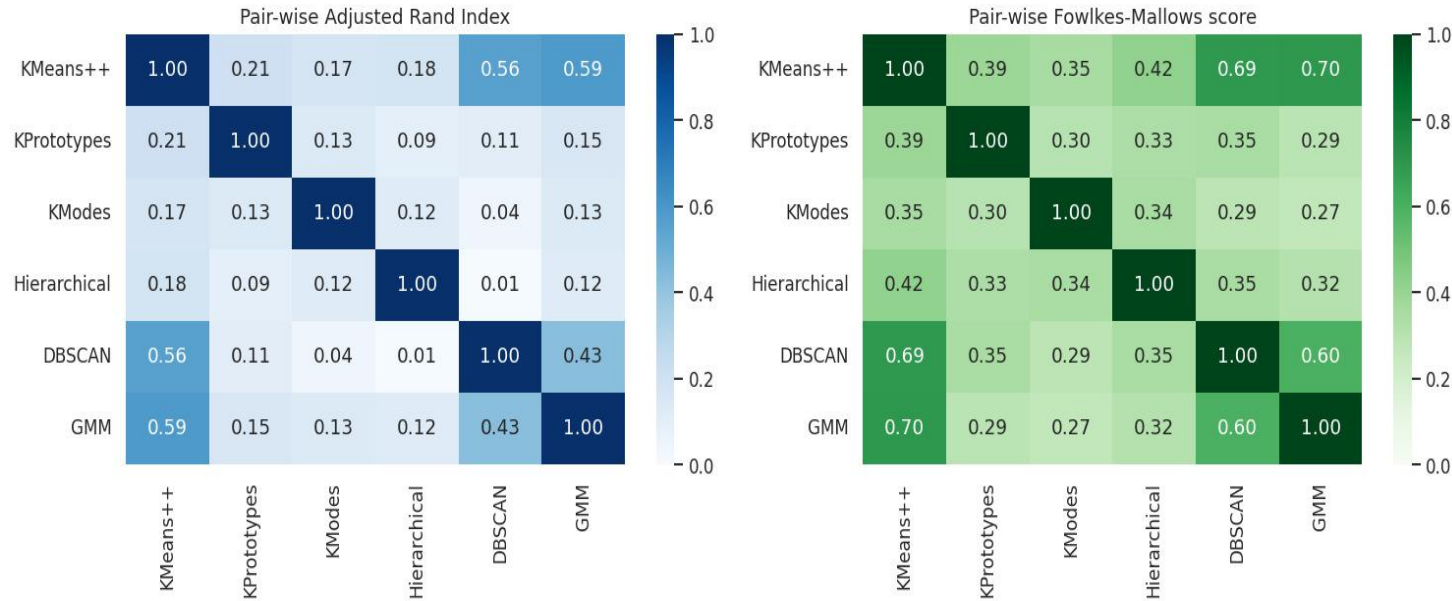


Fig. 10 — Adjusted Rand Index (left) and Fowlkes-Mallows score (right) for every pair of clusterings.

Distance > algorithm

The choice of dissimilarity shapes the partition more than the optimisation family.

Euclidean cluster

K-Means++ $\Leftrightarrow$ GMM FM = 0.70

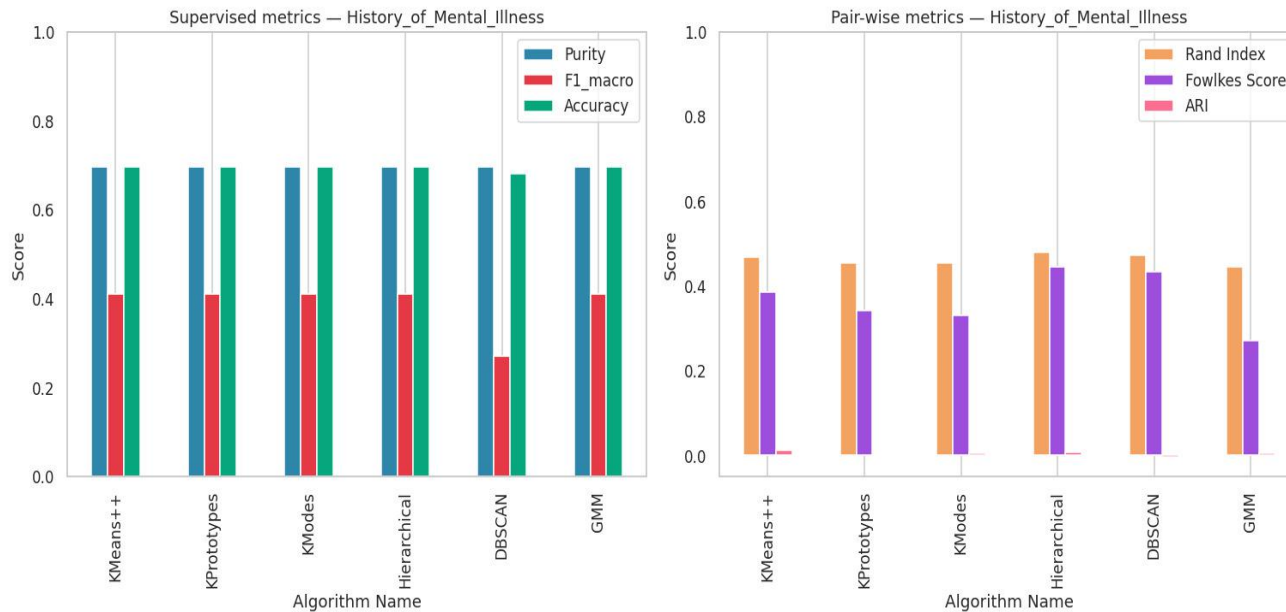
K-Means++ $\Leftrightarrow$ DBSCAN FM = 0.69

Matching / Gower cluster

K-Modes & Hierarchical score FM < 0.35 against the Euclidean group — structurally different partitions.

Take-away — three parallel views capture genuinely different structure; an unsupervised study without type-aware encoding would have missed it.

Supervised metrics vs. History of Mental Illness



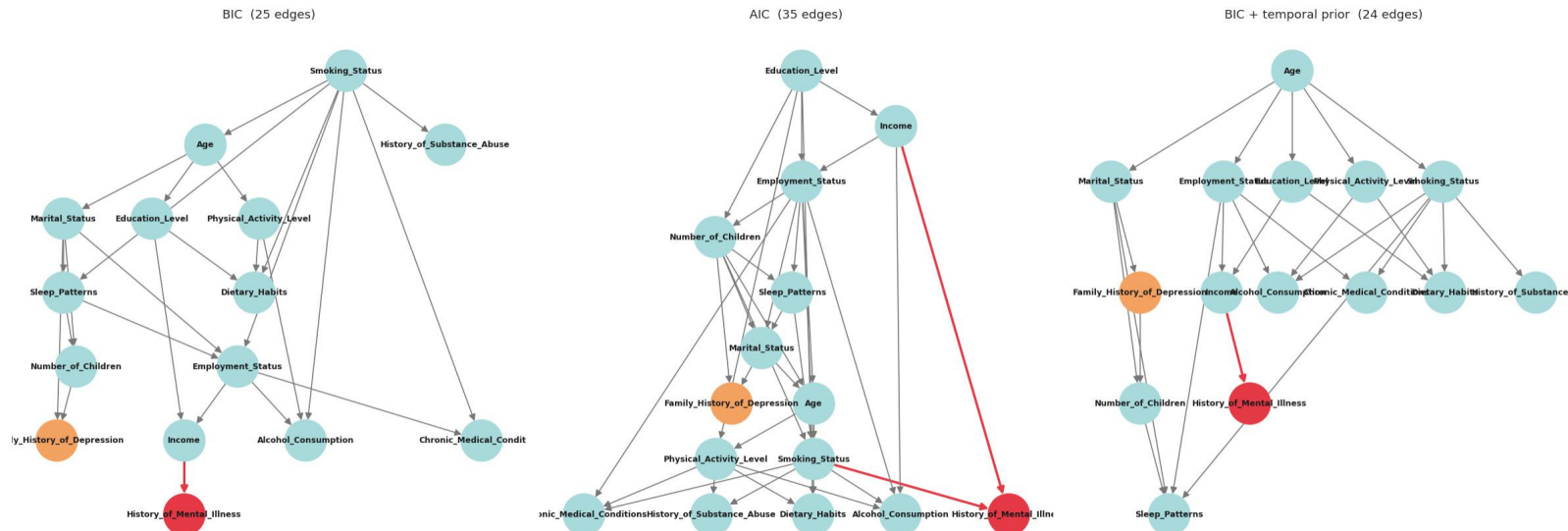
Algorithm	K	Purity	F1	RI	FM
K-Means++	5	0.696	0.410	0.585	0.585
K-Prototypes	5	0.696	0.433	0.587	0.585
K-Modes	6	0.696	0.410	0.583	0.583
Hierarchical	3	0.696	0.411	0.580	0.580
DBSCAN	8	0.696	0.272	0.585	0.587
GMM	10	0.696	0.410	0.580	0.580

Shared F1 $\approx$ 0.41 ceiling

Every cluster maps to the majority class — the partitions encode generic demographic structure, not disease-specific structure. This motivates the move to a Bayesian Network.

Fig. 11a — Purity, F1-macro, Rand Index and Fowlkes–Mallows for all six clusterings.

Structure & parameter learning



Hill-Climbing DAGs under BIC (25 edges), AIC (35 edges), and BIC + temporal expert prior (24 edges).

Markov blanket of the target

BIC $\rightarrow$ {Income} **AIC** $\rightarrow$ {Income, Smoking_Status} **BIC + temporal** $\rightarrow$ {Income}

Hill-Climbing search

on X_{cat} with BIC · AIC · BIC + temporal prior.

Parameters

MLE vs. BDeu (eq. sample size 5). BDeu adopted — strictly positive CPDs even on rare configurations.

Inference

Variable Elimination for exact queries.

Probabilistic Inference & Classifier

Exact Variable Elimination on BDeu-fitted CPDs

Evidence	P(MI = Yes)
Marginal	0.304
Family History = Yes	0.305
Sleep = Poor & Substance Abuse = Yes	0.315
Sleep = Good · Diet = Healthy · Active	0.254

Table 2 · Posterior under selected evidence

Bayesian Network — Confusion (acc 0.69, F1 0.41)

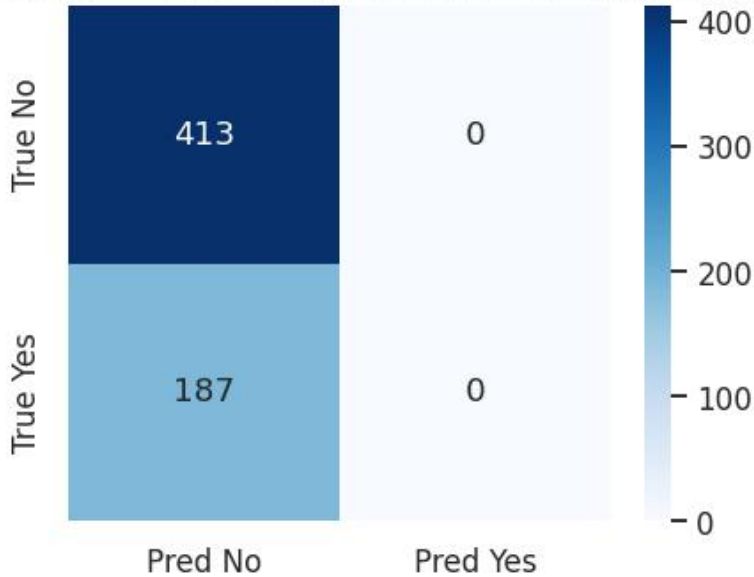
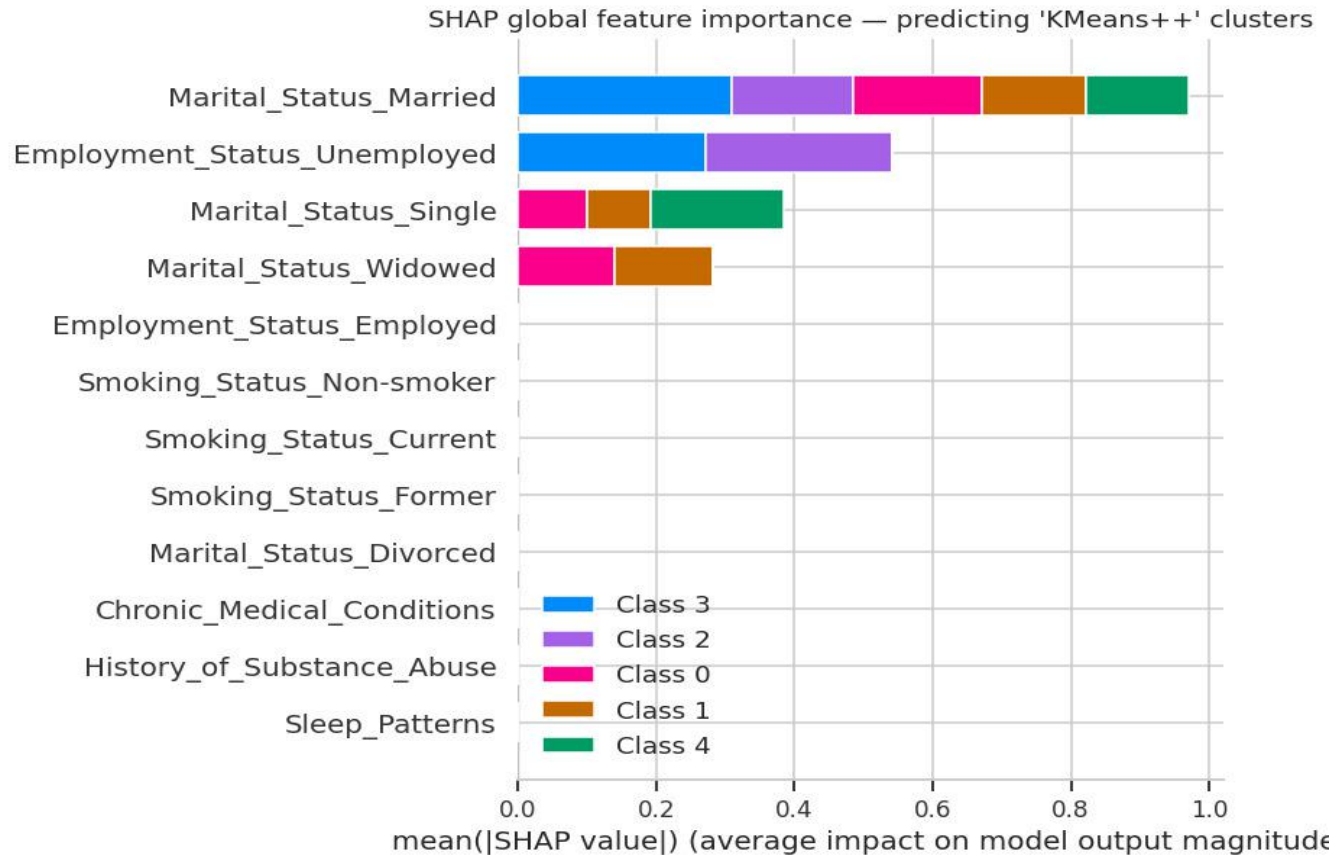


Fig 20 · BN classifier confusion matrix (600 held-out)

BN as classifier Accuracy 0.688 · F1-macro 0.408 — matches the best clustering, but additionally yields calibrated probabilities and partial-evidence inference.

SHAP on a decision-tree surrogate of K-Means++



Convergence across methods

SHAP top features and the BN Markov blanket describe the same socio-economic core.

- ♦ **Marital_Status**
- ♦ **Employment_Status**
- ♦ **Income**

Two independent methodologies converge on the same risk / protective variables — a strong sanity check consistent with the public-health literature on depression.

Fig. 15 — Global feature importance for predicting K-Means++ cluster labels.

Limitations & future work

Limitations

- 1 Low clustering tendency / weak structure — Hopkins ≈ 0.76 is okay but the data isn't strongly clusterable; clusters overlap heavily.
- 2 Supervised ceiling is low — F1-macro ≈ 0.41 , so clusters only weakly align with the target label (predictive usefulness is limited).
- 3 Mixed-type data trade-offs — every encoding (X_num, X_mix/Gower, X_cat) loses information; results depend on the encoding choice.
- 4 Bayesian Network constraints — Hill-Climbing with BIC/AIC finds a local optimum (not guaranteed global); DAG is correlational, not causal; BDeu priors and the temporal prior bias the structure; discretization/binning loses signal.
- 5 No external validation — evaluation is in-sample; no held-out or temporal validation of clusters or BN.

Future work

Causal discovery

Move from association to causation with PC / GES algorithms and, ideally, interventional data.

Richer features

Bring in temporal, clinical and biometric signals beyond self-reported lifestyle variables.

Deep / hybrid clustering

Auto-encoder embeddings + density-based clustering on mixed health data.

Calibrated risk model

Wrap the Bayesian Network into a deployable risk-prediction tool with personalised inference.

Fairness audit

Test partitions and BN posteriors for demographic bias before any clinical use.

Thank you